

EXTENDED DYNAMIC MODE DECOMPOSITION WITH FOURIER DICTIONARIES: ERROR BOUNDS AND FAST IMPLEMENTATION

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ABSTRACT. The Koopman operator has gained considerable attention in dynamical systems due to its capability to provide a linear viewpoint for nonlinear systems using data-driven methods such as extended dynamic mode decomposition (EDMD). In this work, we suggest an EDMD-variant with a Fourier dictionary on the d -dimensional torus, where the data are sampled on an equispaced tensor grid. In this setting, the EDMD regression problem admits a unique closed-form solution, which we show to coincide with trigonometric interpolation of the Koopman image. This identification has two consequences. First, using Koopman invariance of Sobolev spaces, we transfer approximation-theoretic results for trigonometric interpolation to derive error bounds on approximations of the Koopman operator. In particular, the established bounds are of optimal order with explicit constants. Second, the EDMD matrix never has to be assembled, since its action reduces to (nonequispaced) fast Fourier transforms such that the EDMD-approximation may be evaluated matrix-free with quasi-linear cost in the dictionary size. We illustrate the results for the Kuramoto model of coupled oscillators with dictionaries of up to 10^7 modes.

1. INTRODUCTION

The Koopman operator lifts a nonlinear dynamical system to a linear, but infinite-dimensional, operator acting on observable functions [11, 16]. Since linear objects are amenable to spectral analysis along with tailored stable long-term prediction and control [23], this viewpoint has become a central tool in data-driven modeling, see the survey [3] and the collection [15]. To obtain a computable object, one restricts the operator to a finite-dimensional dictionary of observables and fits its action from data snapshots by least squares, which is the essence of extended dynamic mode decomposition (EDMD) [26]. From a computational perspective, EDMD requires the solution of a regression problem whose cost usually grows at least quadratically in the dictionary size, since the Gram matrix of the dictionary on the data has to be formed. This may render EDMD-based methods challenging for high-dimensional problems.

In the infinite-data limit, EDMD converges to the L^2 -orthogonal projection (compression) of the Koopman operator onto the dictionary [12]. Quantitative error bounds for EDMD are comparatively recent. For kernel EDMD [9, 27], that is, EDMD with the dictionary spanned by a reproducing kernel, deterministic L^∞ -bounds in terms of the fill distance of the data were established in [2, 10, 19]. L^2 -type finite-data bounds on the estimation error in terms of the number of snapshots were derived in [18, 20] for a fixed dictionary and in [28] for finite-element dictionaries. All of these bounds rest on approximation results for the respective dictionary, that is, on kernel interpolation estimates in native

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spaces in the L^∞ -case and on finite-element approximation combined with sampling recovery in the L^2 -case. In this work, we build upon a third well-established branch of approximation theory that comes with efficient computations, namely trigonometric approximation of periodic functions.

On the torus $\mathbb{T}^d$, trigonometric interpolation on an equispaced grid of $(2n+1)^d$ points approximates a function of Sobolev smoothness s with an L^2 error of order n^{-s} , which is the best possible rate among all algorithms using this number of point evaluations, see [6, 25] and, for sharp constants of the underlying Sobolev embeddings, [13]. Therein, the interpolation operator does not use any smoothness information and hence adapts automatically to the regularity of the function. This is in contrast to kernel methods, where the kernel usually has to be adapted to obtain a faster convergence rate for higher-order Sobolev spaces. Moreover, on the torus the fractional Sobolev spaces coincide *isometrically* with the interpolation spaces between $L^2(\mathbb{T}^d)$ and an integer-order Sobolev space [4], such that the approximation results for integer smoothness directly transfer to all intermediate smoothness orders without introducing any additional interpolation constants. On the computational side, the evaluation matrix of a Fourier dictionary on an equispaced grid is the d -dimensional discrete Fourier transform matrix, which is unitary up to scaling allowing for fast inversion which can be evaluated by means of the fast Fourier transform (FFT) in quasi-linear time [5, 21]. Evaluating trigonometric polynomials at arbitrary, nonequispaced points is possible at the same cost by the nonequispaced FFT (NFFT) [1, 8, 21]. We use both transforms: the FFT to pass between the values of an observable on the grid and its Fourier coefficients, and the NFFT to evaluate the dictionary functions at the image points $F(x)$ of the dynamics F , which are in general not equispaced.

This work suggests the implementation of Fourier-based techniques for fast EDMD with error guarantees. To this end, we choose a Fourier dictionary of bandwidth n on $\mathbb{T}^d$ and sample the dynamics on the equispaced grid of $(2n+1)^d$ points, such that the number of data points equals the dimension of the dictionary. The key observation is that the EDMD regression problem then has a unique solution in closed form, and the inverse appearing therein is, up to scaling, the adjoint of a Fourier matrix, such that its application can be realized by means of the FFT. We will show that the resulting EDMD matrix is precisely the matrix representation of trigonometric interpolation composed with the Koopman operator, that is, EDMD with Fourier dictionaries on equispaced grids coincides with trigonometric interpolation of the Koopman image. This identity enables an L^2 error analysis, since interpolation estimates transfer directly to the Koopman approximation.

Fourier dictionaries have been used in Koopman theory and EDMD before, but with different objectives. In [7], Fourier bases and the FFT are used to approximate Koopman spectra of measure-preserving maps on the torus, where the focus lies on spectral convergence rather than on quantitative operator error bounds. Furthermore, [17] uses random Fourier features [22] to reduce the computational cost of kernel EDMD and interprets the resulting method as EDMD with a Fourier dictionary with randomly drawn frequencies. In contrast to these approaches, we exploit a deterministic, structured Fourier dictionary together with its corresponding equispaced grid, which enables both FFT-based computation and a priori error estimates.

Contribution. This paper features two main contributions, the first being the closed-form solution of EDMD with Fourier dictionaries together with its matrix-free implementation based on the FFT and the NFFT, the second being error bounds of the form $\|\mathcal{K} - \hat{\mathcal{K}}\|_{H^s(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)} \leq Cn^{-s}$ for two distinct Koopman approximants, one for the case that the observable is measured along the dynamics and one for the fully data-driven case that only its values on the grid are available. The rate is optimal and the constant C is explicit in terms of the map F , its smoothness, the dimension d and the smoothness s of the observable, such that the bound can be evaluated a priori. As by-products we obtain an error estimate for trigonometric interpolation in Sobolev spaces with explicit constants, boundedness of the Koopman operator on Sobolev spaces on periodic domains of fractional order, and the isometric identification of these spaces with interpolation spaces.

Outline. The paper is organized as follows. In Section 2 we recall EDMD and derive the closed-form solution for Fourier dictionaries on equispaced grids. In Section 3 we identify this solution with trigonometric interpolation and introduce the two Koopman approximants. Section 4 contains the error analysis, and Section 5 the numerical experiments for the Kuramoto model. Section 6 concludes the paper.

2. EXTENDED DYNAMIC MODE DECOMPOSITION WITH FOURIER DICTIONARIES

Let $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ be the d -dimensional torus represented in the Euclidean space by $\mathbb{T}^d = [0, 1]^d$ with opposite faces identified. We consider a discrete time dynamical system

$$x^+ = F(x)$$

with $F : \mathbb{T}^d \rightarrow \mathbb{T}^d$ a continuous, potentially nonlinear map. Then F induces the linear *Koopman operator* $\mathcal{K}$ mapping functions on $\mathbb{T}^d$ to functions on $\mathbb{T}^d$ via

$$\mathcal{K}f := f \circ F, \quad f : \mathbb{T}^d \rightarrow \mathbb{C}. \quad (2.1)$$

The key observation is that, after the application of the Koopman operator, we follow the evolution of observables $f : \mathbb{T}^d \rightarrow \mathbb{C}$ instead of the state and that this propagation is linear even if F is nonlinear. The tradeoff is that the Koopman-lifted system is, in general, infinite-dimensional.

2.1. EDMD. We briefly recall *Extended Dynamic Mode Decomposition* (EDMD; [26]), a widely-used data-driven method to approximate the Koopman operator. Assume that we are given $M \in \mathbb{N}$ data snapshots (x_i, y_i) with $y_i = F(x_i)$, $i = 1, \dots, M$, and define the sets

$$X = \{x_1, \dots, x_M\} \quad \text{and} \quad Y = \{y_1, \dots, y_M\}.$$

Choosing a set of linearly independent observables $\psi_k : \mathbb{T}^d \rightarrow \mathbb{C}$, $k = 1, \dots, N$, we define the finite-dimensional space, often referred to as *dictionary*, by

$$\mathbb{V}_N := \text{span}\{\psi_1, \dots, \psi_N\}.$$

Directly from the Koopman identity (2.1), we obtain

$$\mathcal{K}\psi_k(x_j) = \psi_k(F(x_j)) = \psi_k(y_j).$$

Therefore, EDMD seeks a matrix $K \in \mathbb{C}^{N \times N}$ such that

$$\Psi_Y \approx K \Psi_X \quad (2.2)$$

where $\Psi(x) = (\psi_1(x), \dots, \psi_N(x))^T$ collects the evaluation of the dictionary functions at the data snapshots and

$$\Psi_X = [\Psi(x_1), \dots, \Psi(x_M)], \quad \Psi_Y = [\Psi(y_1), \dots, \Psi(y_M)] \in \mathbb{C}^{N \times M}.$$

The approximation (2.2) is then performed by solving the least squares problem

$$\min_{K \in \mathbb{C}^{N \times N}} \|K \Psi_X - \Psi_Y\|_F^2 = \min_{K \in \mathbb{C}^{N \times N}} \sum_{i=1}^M \|K \Psi(x_i) - \Psi(y_i)\|_2^2. \quad (2.3)$$

Here, the Frobenius norm corresponds to the empirical L^2 -norm induced by the sampled data. Consequently, the minimizer yields the best least-squares fit of the Koopman action on the dictionary at the sample points. Assuming that Ψ_X has full row rank, the solution of the least squares problem is given by

$$K = \Psi_Y \Psi_X^\dagger = \Psi_Y \Psi_X^* (\Psi_X \Psi_X^*)^{-1}. \quad (2.4)$$

This matrix K represents a Monte-Carlo-type data-based approximation of the action of the Koopman operator restricted to the dictionary basis $\{\psi_1, \dots, \psi_N\}$. As shown in [12], under suitable assumptions,

this approximation converges to the L^2 -orthogonal projection $P_N \mathcal{K}|_{\mathbb{V}_N}$ as the number of data points tends to infinity.

2.2. EDMD with Fourier dictionaries on equispaced grids. In this work, we show how Fourier dictionaries are particularly well-suited for fast and provably optimal approximations in terms of the number of samples. For a multi-index $k = (k_1, \dots, k_d)^\top \in \mathbb{Z}^d$ we define the Fourier basis functions

$$\psi_k(x) = \exp(2\pi i \langle k, x \rangle), \quad x \in \mathbb{T}^d. \quad (2.5)$$

where $\langle k, x \rangle := \sum_{j=1}^d k_j x_j$. The family $\{\psi_k\}_{k \in \mathbb{Z}^d}$ forms an orthonormal basis of $L^2(\mathbb{T}^d)$. For a fixed bandwidth $n \in \mathbb{N}$ we consider the finite dictionary

$$\mathbb{V}_n = \text{span}\{\psi_k : k \in K_n\}, \quad K_n := \{k \in \mathbb{Z}^d : \|k\|_\infty \leq n\}. \quad (2.6)$$

Thus, we have $\dim(\mathbb{V}_n) = (2n+1)^d =: N$. To compute the approximation via EDMD, we consider a tensor-product equispaced sampling grid given by

$$X = \left\{ x_j = \left(\frac{j_1}{2n+1}, \dots, \frac{j_d}{2n+1} \right) : j_1, \dots, j_d = 0, \dots, 2n \right\} \subset \mathbb{T}^d. \quad (2.7)$$

Therein, the points in X are indexed by multi-indices $j \in \mathcal{I}_n$ with $\mathcal{I}_n := \{0, \dots, 2n\}^d$. Importantly, the number of $N = (2n+1)^d$ grid points matches the dimension of the Fourier dictionary, that is, $|X| = \dim(\mathbb{V}_n)$.

Given data snapshots $(x_j, F(x_j))$ for $x_j \in X, j \in \mathcal{I}_n$ and collecting the evaluation of the dictionary functions at the data snapshots in the $N \times N$ matrices

$$(L_X)_{k,j} = \psi_k(x_j) \quad \text{and} \quad (L_{X+})_{k,j} = \psi_k(F(x_j)),$$

where rows are indexed by frequencies $k \in K_n$ and columns by data points $j \in \mathcal{I}_n$, the EDMD least squares problem (see (2.3)) becomes

$$\min_{K \in \mathbb{C}^{N \times N}} \|KL_X - L_{X+}\|_F^2.$$

Due to the particular structure of the dictionary and the sample points, the matrix L_X has favorable properties allowing for fast inversion that lay the foundation for fast Fourier methods. This is a consequence of the orthogonality of the Fourier basis on the equispaced grid: by [21, Lemma 4.66], we have for every $m \in \mathbb{Z}^d$

$$\frac{1}{N} \sum_{j \in \mathcal{I}_n} \psi_m(x_j) = \begin{cases} 1, & m \in (2n+1)\mathbb{Z}^d, \\ 0, & \text{otherwise.} \end{cases} \quad (2.8)$$

Lemma 1. *Let X be the uniform tensor-product grid defined in (2.7). Then*

$$L_X L_X^* = N I_N$$

such that L_X is invertible with

$$L_X^{-1} = \frac{1}{N} L_X^*.$$

Proof. For $k, \ell \in K_n$ we have $(L_X L_X^*)_{k,\ell} = \sum_{j \in \mathcal{I}_n} \psi_{k-\ell}(x_j)$, and $\|k - \ell\|_\infty \leq 2n < 2n+1$ implies that $k - \ell \in (2n+1)\mathbb{Z}^d$ if and only if $k = \ell$. Hence (2.8) yields $L_X L_X^* = N I_N$. $\blacksquare$

Consequently, using a Fourier dictionary and equispaced points, the inverse entering the solution of the EDMD regression problem is known analytically and therefore never has to be computed. In particular, it has a very simple form and is expressed only using the data matrices L_X and L_{X+} .

Corollary 2. *Let $\mathbb{V}_n$ be the Fourier dictionary and let X be the equispaced tensor-product grid defined in (2.7). Denote by L_X, L_{X+} the corresponding Fourier evaluation matrices. Then the EDMD least-squares problem admits the unique solution*

$$K = L_{X+} L_X^{-1} = \frac{1}{N} L_{X+} L_X^*. \quad (2.9)$$

The explicit representation of the EDMD matrix K obtained in Corollary 2 is the key structural property of Fourier-EDMD on equispaced grids. It has two fundamental consequences that constitute the core of this work.

First, $L_X^{-1} = \frac{1}{N} L_X^*$ admits an interpretation as reconstruction of trigonometric polynomials from their values on the sampling grid. In particular, the operator associated with K can be interpreted as the composition which maps sampled data to reconstructed Fourier representations after transport by the dynamics F . As we show in Section 3, this leads to an exact characterization of Fourier-EDMD in terms of trigonometric interpolation, which allows us to transfer classical results from approximation theory to the analysis of EDMD. To be precise, the matrix K induces a linear operator

$$\mathcal{K}_n : \mathbb{V}_n \rightarrow \mathbb{V}_n, \quad (\mathcal{K}_n f)(x) = (K^\top \tilde{f})^\top \Psi(x) = \tilde{f}^\top K \Psi(x)$$

where $\tilde{f}$ denotes the coefficient vector of $f \in \mathbb{V}_n$, $f(x) = \tilde{f}^\top \Psi(x)$. The operator $\mathcal{K}_n$ represents the finite-dimensional Fourier-EDMD approximation of the Koopman operator $\mathcal{K}$ on the space $\mathbb{V}_n$. In Section 4 it will be shown that, under suitable Sobolev regularity assumptions, the approximation error satisfies an error bound of the form

$$\|(\mathcal{K} - \mathcal{K}_n)f\|_{L^2(\mathbb{T}^d)} \leq C n^{-s} \|f\|_{H^s(\mathbb{T}^d)}, \quad \forall f \in \mathbb{V}_n$$

where the constant $C > 0$ is explicitly known. Using a suitable projection from the right, we will further show that a similar estimate also holds for functions in $H^s(\mathbb{T}^d)$. Here, we stress that the rate n^{-s} is optimal in the worst-case setting for observables with Sobolev smoothness s and $(2n+1)^d$ point evaluations.

Second, the matrix L_X possesses a discrete Fourier transform structure. Consequently, both L_X and L_X^{-1} and hence the EDMD approximation K can be applied by means of multidimensional FFTs and NFFT, allowing for highly-efficient matrix-free implementations with quasi-linear complexity $\mathcal{O}(N \log N)$ with the total number of points and dictionary size $N = (2n+1)^d$ that enables computations for high-dimensional dictionaries, as illustrated in Section 5.

3. EDMD AS TRIGONOMETRIC INTERPOLATION

Corollary 2 shows that the Fourier-EDMD approximation is completely determined by the Fourier sampling matrices L_X and L_{X+} . The purpose of this section is to show that Fourier-EDMD on equispaced grids coincides with trigonometric interpolation of Koopman images. This identification forms the basis for the error analysis performed in Section 4.

3.1. Trigonometric interpolation on equispaced grids. We first recall the trigonometric interpolation operator associated with the equispaced grid X .

Definition 3. *Let X be the equispaced tensor-product grid (2.7). The trigonometric interpolation operator*

$$S_n : C(\mathbb{T}^d) \rightarrow \mathbb{V}_n$$

is defined by assigning to every $f \in C(\mathbb{T}^d)$ the unique trigonometric polynomial $S_n f \in \mathbb{V}_n$ satisfying

$$(S_n f)(x_j) = f(x_j), \quad j \in \mathcal{I}_n.$$

To provide a relation of the trigonometric interpolation operator S_n with the matrix L_X involved in the EDMD surrogate, let

$$\Psi(x) := (\psi_k(x))_{k \in K_n}.$$

Then, by definition, every $p \in \mathbb{V}_n$ admits a unique representation $p(x) = c^\top \Psi(x)$ for a coefficient vector $c = (c_k)_{k \in K_n} \in \mathbb{C}^N$. For a continuous function $f \in C(\mathbb{T}^d)$ we denote by

$$f_X = (f(x_j))_{j \in \mathcal{I}_n} \in \mathbb{C}^N$$

the vector of its values on the interpolation grid.

Lemma 4. *Let $f \in C(\mathbb{T}^d)$ and let $S_n f(x) = \tilde{f}^\top \Psi(x) \in \mathbb{V}_n$ be the trigonometric interpolant of f with coefficient vector $\tilde{f} \in \mathbb{C}^N$. Then, the coefficients satisfy*

$$\tilde{f}^\top = \frac{1}{N} f_X^\top L_X^*.$$

In particular, $S_n f$ admits the representation

$$S_n f(x) = \frac{1}{N} f_X^\top L_X^* \Psi(x).$$

Proof. For every $j \in \mathcal{I}_n$ we have

$$S_n f(x_j) = \tilde{f}^\top \Psi(x_j) = \tilde{f}^\top (L_X)_{:,j}$$

since the j -th column of L_X is $\Psi(x_j)$. Collecting these values for all $j \in \mathcal{I}_n$ yields

$$(S_n f)_X^\top = \tilde{f}^\top L_X.$$

Moreover, since $S_n f$ interpolates f on the grid X , we have

$$f_X^\top = (S_n f)_X^\top = \tilde{f}^\top L_X$$

and hence

$$\tilde{f}^\top = f_X^\top (L_X)^{-1} = \frac{1}{N} f_X^\top L_X^*$$

where the second equality holds by Lemma 1. ■

The representation above provides an explicit characterization of trigonometric interpolation in terms of the Fourier evaluation matrix. The matrix L_X^{-1} maps function values on the grid to Fourier coefficients, while the feature vector $\Psi(x)$ evaluates the resulting trigonometric polynomial at arbitrary points. Next, we show that the EDMD approximation obtained in Section 2 coincides with trigonometric interpolation applied to Koopman images.

3.2. The EDMD matrix as trigonometric interpolation. First, we show that the EDMD approximant preserves the fundamental property of the Koopman operator propagating observable functions along the flow.

Lemma 5. *For every sampling point $x_j \in X$, $j \in \mathcal{I}_n$,*

$$K\Psi(x_j) = \Psi(F(x_j)). \tag{3.1}$$

Proof. By Corollary 2, that is, right-multiplication of (2.9) with L_X , we have $KL_X = L_{X^+}$ such that the result follows. ■

Lemma 5 shows that the EDMD matrix reproduces the action of the dynamical system on the feature vectors at the sampling points. In other words, the finite-dimensional relation (3.1) is the discrete analogue of the Koopman identity

$$(\mathcal{K}f)(x) = f(F(x)).$$

Consequently, the EDMD approximation may be interpreted as propagating Fourier feature vectors through the dynamics and subsequently reconstructing a trigonometric polynomial from the propagated data.

Proposition 6. *Let $f(x) = \tilde{f}^\top \Psi(x) \in \mathbb{V}_n$. Then*

$$(S_n \mathcal{K}f)(x) = \tilde{f}^\top K \Psi(x),$$

i.e. the matrix K represents the operator $S_n \mathcal{K}|_{\mathbb{V}_n}$ with respect to the Fourier basis of $\mathbb{V}_n$, see (2.6).

Proof. Let $f \in \mathbb{V}_n$ with $f(x) = \tilde{f}^\top \Psi(x)$. By Lemma 5,

$$\tilde{f}^\top K \Psi(x_j) = \tilde{f}^\top \Psi(F(x_j)) = f(F(x_j))$$

for every sampling point $x_j \in X$. Hence the trigonometric polynomial

$$p(x) := \tilde{f}^\top K \Psi(x)$$

interpolates the function $f \circ F$ on the grid X . Since $p \in \mathbb{V}_n$ and as the trigonometric interpolant is unique,

$$p = S_n(f \circ F) = S_n \mathcal{K}f,$$

which proves the claim. ■

Proposition 6 shows that EDMD with the particular choice of a Fourier dictionary and an equispaced grid is not merely a least-squares approximation procedure. On the dictionary space $\mathbb{V}_n$, it coincides with trigonometric interpolation of the Koopman image of the dictionary. This interpretation naturally extends to arbitrary continuous observables and leads to two Koopman approximants, depending on the available data. These approximants were already introduced and discussed in the context of kernel EDMD in [10], see also [2, Remark 2.3].

Firstly, if the values $f(F(x_j))$ are available we may consider the Koopman approximant

$$\hat{\mathcal{K}}_1 := S_n \mathcal{K} : C(\mathbb{T}^d) \rightarrow \mathbb{V}_n \quad (3.2)$$

which maps an observable $f \in C(\mathbb{T}^d)$ to the trigonometric interpolant of $f \circ F$, i.e.,

$$\hat{\mathcal{K}}_1 f = S_n(f \circ F) = \frac{1}{N} (f \circ F)_X^\top L_X^* \Psi.$$

This approximant may be implemented using the fast Fourier transform.

If only the values on the *original* sampling grid X are available, one must first replace f by its trigonometric interpolant. This leads to the data-driven approximation

$$\hat{\mathcal{K}}_2 := S_n \mathcal{K} S_n : C(\mathbb{T}^d) \rightarrow \mathbb{V}_n. \quad (3.3)$$

Then, the Koopman approximant (3.3) applied to f is

$$\hat{\mathcal{K}}_2 f = S_n \mathcal{K} S_n f = S_n \mathcal{K}|_{\mathbb{V}_n} (S_n f) = \widetilde{(S_n f)}^\top K \Psi = \tilde{f}^\top K \Psi = \frac{1}{N^2} f_X^\top L_X^* L_{X+} L_X^* \Psi.$$

Note that the application of this operator only requires the values of f on the grid X , together with the evaluation of the dictionary functions at the image points $F(x_j)$, but does not require direct evaluation of f at off-grid locations. While the multiplication with L_X^* can be carried out via FFT, the multiplication with L_{X+} involves evaluations of a trigonometric polynomial at the nonuniform points $F(x_j)$ and therefore requires an extension of the FFT to nonequispaced points (NFFT).

4. ERROR ANALYSIS IN SOBOLEV SPACES

The equivalence of Fourier-EDMD and trigonometric interpolation of Koopman images established in the previous section implies that the EDMD approximation error in Sobolev spaces is governed by two ingredients. First, the approximation properties of trigonometric interpolation, and, second, the invariance of Sobolev spaces under the Koopman operator together with a bound on its operator norm. In this section, we quantify both and combine them into an error bound with explicit constants.

4.1. Sobolev spaces on the torus. We first introduce Sobolev spaces on the torus used throughout this work. For $f \in L^2(\mathbb{T}^d)$ let $\hat{f}_k = \langle f, \psi_k \rangle_{L^2(\mathbb{T}^d)} = \int_{\mathbb{T}^d} f(x) \exp(-2\pi i \langle k, x \rangle) dx$, $k \in \mathbb{Z}^d$, denote its Fourier coefficients, i.e. $f = \sum_{k \in \mathbb{Z}^d} \hat{f}_k \psi_k$. For $s \geq 0$ we set

$$\|f\|_{H^s(\mathbb{T}^d)}^2 := \sum_{k \in \mathbb{Z}^d} (1 + \|k\|_2^2)^s |\hat{f}_k|^2 \quad (4.1)$$

and define the *fractional* Sobolev space

$$H^s(\mathbb{T}^d) := \left\{ f \in L^2(\mathbb{T}^d) : \|f\|_{H^s(\mathbb{T}^d)} < \infty \right\}. \quad (4.2)$$

In particular, $H^0(\mathbb{T}^d) = L^2(\mathbb{T}^d)$ isometrically.

It is well-known that for integer smoothness $\sigma \in \mathbb{N}$, this norm is equivalent to the derivative-based Sobolev norm

$$\|f\|_{H_D^\sigma(\mathbb{T}^d)}^2 = \sum_{\|\alpha\|_1 \leq \sigma} \|\partial^\alpha f\|_{L^2(\mathbb{T}^d)}^2, \quad \partial^\alpha f = \frac{\partial^{\|\alpha\|_1} f}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}.$$

This norm equivalence, with explicit dimension-independent constants, is established for the torus $[0, 2\pi]^d$ in [13, eq. (2.3)] with a slightly different upper bound. We restate it in the scaling of the torus used in this work to explicitly control constants.

Lemma 7. *Let $\sigma \in \mathbb{N}$. Then for all $f \in H^\sigma(\mathbb{T}^d)$*

$$\frac{1}{\sqrt{\sigma!}} \|f\|_{H^\sigma(\mathbb{T}^d)} \leq \|f\|_{H_D^\sigma(\mathbb{T}^d)} \leq (2\pi)^\sigma \|f\|_{H^\sigma(\mathbb{T}^d)}.$$

Proof. We follow the proof of [13, eq. (2.3)]. By the multinomial identity, we have

$$(1 + \|k\|_2^2)^\sigma = \sum_{\|\alpha\|_1 \leq \sigma} \underbrace{\frac{\sigma!}{(\sigma - \|\alpha\|_1)! \alpha_1! \cdots \alpha_d!}}_{=: c_\alpha} \prod_{i=1}^d k_i^{2\alpha_i}, \quad (4.3)$$

with $1 \leq c_\alpha \leq \sigma!$. Since $\partial^\alpha \psi_k = \prod_{i=1}^d (2\pi i k_i)^{\alpha_i} \psi_k$, Parseval's identity gives

$$\|f\|_{H_D^\sigma(\mathbb{T}^d)}^2 = \sum_{k \in \mathbb{Z}^d} w(k) |\hat{f}_k|^2 \quad \text{with} \quad w(k) := \sum_{\|\alpha\|_1 \leq \sigma} \prod_{i=1}^d (2\pi k_i)^{2\alpha_i}.$$

That is, $w(k)$ is the right-hand side of (4.3) with $2\pi k$ in place of k and all coefficients c_α replaced by one. Since all summands are nonnegative, comparing the two sums termwise gives

$$w(k) \leq (1 + 4\pi^2 \|k\|_2^2)^\sigma \leq \sigma! w(k).$$

Together with $1 + \|k\|_2^2 \leq 1 + 4\pi^2 \|k\|_2^2 \leq 4\pi^2 (1 + \|k\|_2^2)$, this yields

$$\frac{1}{\sigma!} (1 + \|k\|_2^2)^\sigma \leq w(k) \leq (2\pi)^{2\sigma} (1 + \|k\|_2^2)^\sigma.$$

and taking square roots proves the claim. ■

4.2. Boundedness of the Koopman operator. The following theorem uses the norm equivalence established above to transfer known bounds for the Koopman operator in the derivative-based Sobolev norm to the Fourier-based Sobolev norm (4.1). To this end, for a multi-index $\gamma \in \mathbb{N}_0^d$ the set $\mathcal{P}(\gamma)$ contains all tuples $\beta = (\beta_1, \dots, \beta_{\|\gamma\|_1}) \in \{1, \dots, d\}^{\|\gamma\|_1}$ in which every index i occurs exactly γ_i times, so that $\partial^\gamma = \partial_{\beta_1} \cdots \partial_{\beta_{\|\gamma\|_1}}$ for each such β . Further, Π_m denotes the set of partitions π of $\{1, \dots, m\}$ into nonempty blocks and $|\pi|$ the number of blocks of a partition $\pi \in \Pi_m$.

Theorem 8. *Let $\sigma \in \mathbb{N}_0$ and assume that the dynamics map satisfies $F \in C^\sigma(\mathbb{T}^d)$ and is a C^1 -diffeomorphism with*

$$c_0^{-1} := \inf_{x \in \mathbb{T}^d} |\det DF(x)| > 0.$$

Then, the Koopman operator $\mathcal{K} : H^\sigma(\mathbb{T}^d) \rightarrow H^\sigma(\mathbb{T}^d)$ is well defined and bounded with

$$\|\mathcal{K}\|_{H^\sigma(\mathbb{T}^d) \rightarrow H^\sigma(\mathbb{T}^d)} \leq (2\pi)^\sigma \sqrt{\sigma!} C_F, \quad C_F := \left(\max \left\{ c_0, \sum_{1 \leq \|\alpha\|_1 \leq \sigma} S_\alpha(F) \right\} \right)^{1/2}$$

and

$$S_\alpha(F) := \sup_{x \in \mathbb{T}^d} \left| \det DF(x) \right|^{-1} \sum_{k=1}^{\|\alpha\|_1} \sum_{\|\alpha'\|_1=k} \left| \sum_{\substack{\pi \in \Pi_{\|\alpha\|_1} \\ |\pi|=k}} \sum_{\beta' \in \mathcal{P}(\alpha')} \partial_{\beta, B_1} F_{\beta'_1}(x) \cdots \partial_{\beta, B_k} F_{\beta'_k}(x) \right|^2$$

for $\alpha \in \mathbb{N}_0^d$, $1 \leq \|\alpha\|_1 \leq \sigma$, and any $\beta \in \mathcal{P}(\alpha)$ where B_i denotes the i -th block of the partition π , and $\partial_{\beta, B}$ stands for the operator $\partial_{\beta_{j_1}} \cdots \partial_{\beta_{j_\ell}}$, where $B = \{j_1, \dots, j_\ell\}$.

Proof. In [10, Theorem 4.2], well-definedness as well as boundedness are shown for the Koopman operator in the derivative-based norm, that is, the bound $\|\mathcal{K}f\|_{H_D^\sigma(\mathbb{T}^d)} \leq C_F \|f\|_{H_D^\sigma(\mathbb{T}^d)}$. The norm equivalence of Lemma 7 implies for the Fourier-based norm

$$\|\mathcal{K}f\|_{H^\sigma(\mathbb{T}^d)} \leq \sqrt{\sigma!} \|\mathcal{K}f\|_{H_D^\sigma(\mathbb{T}^d)} \leq \sqrt{\sigma!} C_F \|f\|_{H_D^\sigma(\mathbb{T}^d)} \leq (2\pi)^\sigma \sqrt{\sigma!} C_F \|f\|_{H^\sigma(\mathbb{T}^d)}. \quad \blacksquare$$

The previous result establishes boundedness of the Koopman operator in $H^\sigma(\mathbb{T}^d)$ for integer smoothness σ . For $\sigma = 0$ the sum in the definition of the constant C_F is empty and the bound reads $\|\mathcal{K}\|_{L^2(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)} \leq \sqrt{c_0}$, which follows from the change of variables $y = F(x)$.

As in [10, Theorem 4.2] we use interpolation theory of Hilbert spaces (see e.g. [4]), to extend these estimates to fractional-order Sobolev spaces. On the torus, the Fourier characterization of Sobolev spaces allows for a particularly simple description as fractional Sobolev spaces coincide *isometrically* with the interpolation spaces between $L^2(\mathbb{T}^d)$ and $H^\sigma(\mathbb{T}^d)$. In the following result, $(\cdot, \cdot)_{\theta, 2}$ denotes the real interpolation space obtained by the K -method with the normalized norm of [4, Section 2].

Lemma 9. *Let $\sigma \in \mathbb{N}$ and $s \in (0, \sigma)$. Then, for $\theta = s/\sigma \in (0, 1)$ we have*

$$H^s(\mathbb{T}^d) = (L^2(\mathbb{T}^d), H^\sigma(\mathbb{T}^d))_{\theta, 2} \quad (4.4)$$

with equality of norms.

Proof. Consider the measure space $(\mathbb{Z}^d, \mathcal{P}(\mathbb{Z}^d), \mu)$ with counting measure μ . Denote by the mapping $\mathcal{F}, f \mapsto (\hat{f}_k)_{k \in \mathbb{Z}^d}$ the Fourier transform and set $w_0(k) \equiv 1$ and $w_1(k) = (1 + \|k\|_2^2)^\sigma$. By Parseval's identity

$$\|f\|_{L^2(\mathbb{T}^d)}^2 = \sum_{k \in \mathbb{Z}^d} |\hat{f}_k|^2 = \|\mathcal{F}f\|_{L^2(\mathbb{Z}^d, w_0\mu)}^2$$

and similarly, by the definition (4.1) of the Sobolev norm,

$$\|f\|_{H^\sigma(\mathbb{T}^d)}^2 = \sum_{k \in \mathbb{Z}^d} (1 + \|k\|_2^2)^\sigma |\widehat{f}_k|^2 = \|\mathcal{F}f\|_{L^2(\mathbb{Z}^d, w_1\mu)}^2.$$

In particular, the mappings $\mathcal{F} : L^2(\mathbb{T}^d) \rightarrow L^2(\mathbb{Z}^d, w_0\mu)$ as well as $\mathcal{F} : H^\sigma(\mathbb{T}^d) \rightarrow L^2(\mathbb{Z}^d, w_1\mu)$ are unitary isomorphisms. Hence, the result follows from [4, Corollary 3.2]. $\blacksquare$

Remark 10. For general domains $\Omega \subset \mathbb{R}^d$, the identification of Sobolev spaces with interpolation spaces typically holds only up to equivalence of norms. More precisely, under suitable regularity assumptions on Ω , e.g. bounded Lipschitz domains, one has

$$H^s(\Omega) = (L^2(\Omega), H^\sigma(\Omega))_{s/\sigma, 2}$$

with equivalent norms. However, the equivalence constants depend on the domains and associated extension operators and are, in general, unknown.

The isometric identification of fractional Sobolev spaces on the torus with interpolation spaces enables us to transfer the operator bounds in Theorem 8 to arbitrary smoothness without introducing additional interpolation constants.

Proposition 11. Let $\sigma \in \mathbb{N}$ and assume that $F \in C^\sigma(\mathbb{T}^d)$ is a C^1 -diffeomorphism satisfying $c_0^{-1} = \inf_{x \in \mathbb{T}^d} |\det DF(x)| > 0$. Then for all $s \in (0, \sigma]$ the linear Koopman operator

$$\mathcal{K} : H^s(\mathbb{T}^d) \rightarrow H^s(\mathbb{T}^d)$$

is well-defined and bounded. In particular, we have

$$\|\mathcal{K}\|_{H^s(\mathbb{T}^d) \rightarrow H^s(\mathbb{T}^d)} \leq (2\pi)^s c_0^{\frac{\sigma-s}{2\sigma}} (\sigma!)^{\frac{s}{2\sigma}} C_F^{s/\sigma}$$

with C_F as defined in Theorem 8.

Proof. Let $\sigma \in \mathbb{N}$ and $s \in (0, \sigma)$. By Lemma 9, $H^s(\mathbb{T}^d) = (L^2(\mathbb{T}^d), H^\sigma(\mathbb{T}^d))_{s/\sigma, 2}$, with equality of norms. Since, by Theorem 8, the Koopman operator $\mathcal{K}$ is well-defined and bounded on $H^\sigma(\mathbb{T}^d)$ and on $L^2(\mathbb{T}^d)$, [24, Lemma 22.3] implies that also $\mathcal{K} : H^s(\mathbb{T}^d) \rightarrow H^s(\mathbb{T}^d)$ is well-defined and bounded with

$$\|\mathcal{K}\|_{H^s(\mathbb{T}^d) \rightarrow H^s(\mathbb{T}^d)} \leq \|\mathcal{K}\|_{L^2(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)}^{1-s/\sigma} \|\mathcal{K}\|_{H^\sigma(\mathbb{T}^d) \rightarrow H^\sigma(\mathbb{T}^d)}^{s/\sigma}.$$

Substituting the bounds from Theorem 8 yields the result. $\blacksquare$

4.3. The trigonometric interpolation error. Having established boundedness of the Koopman operator in Sobolev spaces, we now turn to the approximation error induced by the trigonometric interpolation. We first prove the following auxiliary lemma.

Lemma 12. For $d \in \mathbb{N}$ and $s > d/2$ it holds

$$\sum_{k \in \mathbb{Z}^d \setminus \{0\}} \|k\|_2^{-2s} \leq \frac{2^{2d+1}s}{2s-d}.$$

Proof. We order the sum according to nesting ℓ_∞ -shells

$$\sum_{k \in \mathbb{Z}^d \setminus \{0\}} \|k\|_2^{-2s} = \sum_{R=1}^{\infty} \sum_{\|k\|_\infty=R} \|k\|_2^{-2s} \leq \sum_{R=1}^{\infty} R^{-2s} \sum_{\|k\|_\infty=R} 1 \leq \sum_{R=1}^{\infty} R^{-2s} ((2R+1)^d - (2R-1)^d).$$

By the binomial theorem we have

$$(2R+1)^d - (2R-1)^d = \sum_{j=0}^d \binom{d}{j} (2R)^j - \sum_{j=0}^d \binom{d}{j} (2R)^j (-1)^{d-j}$$

$$\begin{aligned}
&= \sum_{j=0}^{d-1} \binom{d}{j} (2R)^j (1 - (-1)^{d-j}) \\
&\leq 2(2R)^{d-1} \sum_{j=0}^{d-1} \binom{d}{j} \leq 2 \cdot 2^d (2R)^{d-1} = 4^d R^{d-1}.
\end{aligned}$$

such that combining the above relations we get

$$\sum_{k \in \mathbb{Z}^d \setminus \{0\}} \|k\|_2^{-2s} \leq 4^d \sum_{R=1}^{\infty} R^{-2s} R^{d-1} = 4^d \sum_{R=1}^{\infty} R^{-2s+d-1}.$$

Set $\beta := 2s - d + 1$, i.e. that the exponent of the last sum equals $-\beta$, and note that $\beta > 1$ by the assumption $s > d/2$. Since $x \mapsto x^{-\beta}$ is monotone decreasing on $[1, \infty)$, we have $R^{-\beta} \leq \int_{R-1}^R x^{-\beta} dx$ for all $R \geq 2$, and the intervals $[R-1, R]$, $R \geq 2$, cover $[1, \infty)$. Hence, summing the above and keeping the summand $R = 1$ separate gives

$$\begin{aligned}
\sum_{k \in \mathbb{Z}^d \setminus \{0\}} \|k\|_2^{-2s} &\leq 4^d \left(1 + \int_1^{\infty} x^{-\beta} dx \right) = 4^d \left(1 + \frac{1}{\beta - 1} \right) = 4^d \left(1 + \frac{1}{2s - d} \right) \\
&= 4^d \frac{2s - d + 1}{2s - d} \leq \frac{2^{2d+1}s}{2s - d},
\end{aligned}$$

where the last inequality holds since $2s - d + 1 \leq 2s$ for $d \geq 1$. ■

The above lemma together with the Fourier characterization of Sobolev spaces leads to the following approximation estimate for trigonometric interpolation. While the convergence rate is classical, see e.g. [25, Theorem 3.6.4], we require an estimate with fully explicit constants, since it is our aim to derive explicit error bounds for the Fourier-EDMD approximation.

We note that the following error bound on the interpolation is optimal in the sense that no algorithm using $(2n + 1)^d$ point evaluations attains a better rate in the worst case over $H^s(\mathbb{T}^d)$, see [6, 25].

Theorem 13. *Let $d, n \in \mathbb{N}$, $s > d/2$, and $0 \leq r \leq s$ be smoothness parameters and S_n the d -dimensional trigonometric interpolation from Definition 3 based on $(2n + 1)^d$ point evaluations. Then*

$$\|\text{Id} - S_n\|_{H^s(\mathbb{T}^d) \rightarrow H^r(\mathbb{T}^d)} \leq 2^{d+1} \sqrt{\frac{s}{2s-d}} (2d)^{r/2} (n^d)^{\frac{r-s}{d}}.$$

In particular, we have for the L^2 worst-case error

$$\sup_{\|f\|_{H^s(\mathbb{T}^d)} \leq 1} \|f - S_n f\|_{L^2(\mathbb{T}^d)} \leq 2^{d+1} \sqrt{\frac{s}{2s-d}} (n^d)^{-\frac{s}{d}}.$$

Further the trigonometric interpolation operator is bounded from $H^s(\mathbb{T}^d)$ into itself, i.e.

$$\|S_n\|_{H^s(\mathbb{T}^d) \rightarrow H^s(\mathbb{T}^d)} \leq 2^{d+2} \sqrt{\frac{s}{2s-d}} (2d)^{s/2}.$$

Proof. Using Parseval's identity, we decompose the error

$$\|f - S_n f\|_{H^r(\mathbb{T}^d)}^2 = \|f - P_n f\|_{H^r(\mathbb{T}^d)}^2 + \|P_n f - S_n f\|_{H^r(\mathbb{T}^d)}^2, \quad (4.5)$$

where $P_n f = \sum_{\|k\|_{\infty} \leq n} \hat{f}_k \exp(2\pi i \langle k, \cdot \rangle)$ is the L^2 -orthogonal projection. The first summand is the error due to projection, which we bound as follows, using $\|k\|_2 \geq \|k\|_{\infty} \geq n + 1$ and $r - s \leq 0$,

$$\|f - P_n f\|_{H^r(\mathbb{T}^d)}^2 = \sum_{\|k\|_{\infty} > n} (1 + \|k\|_2^2)^{r-s} (1 + \|k\|_2^2)^s |\hat{f}_k|^2$$

$$\begin{aligned}
&\leq (1 + (n+1)^2)^{r-s} \sum_{\|k\|_\infty > n} (1 + \|k\|_2^2)^s |\hat{f}_k|^2 \\
&\leq n^{2(r-s)} \|f\|_{H^s(\mathbb{T}^d)}^2 \leq (2d)^r n^{2(r-s)} \|f\|_{H^s(\mathbb{T}^d)}^2.
\end{aligned}$$

The second summand of (4.5) is due to aliasing effects of the approximation $S_n f$. By Lemma 4, its coefficients are $\tilde{f}_k = \frac{1}{N} \sum_{j \in \mathcal{I}_n} f(x_j) \overline{\psi_k(x_j)}$, $k \in K_n$. Since $s > d/2$, the Fourier series of f converges absolutely, so that we may insert it and interchange the order of summation,

$$\tilde{f}_k = \sum_{m \in \mathbb{Z}^d} \hat{f}_m \frac{1}{N} \sum_{j \in \mathcal{I}_n} \psi_{m-k}(x_j) = \sum_{\ell \in \mathbb{Z}^d} \hat{f}_{k+(2n+1)\ell},$$

by (2.8), since $m - k \in (2n+1)\mathbb{Z}^d$ if and only if $m = k + (2n+1)\ell$ for some $\ell \in \mathbb{Z}^d$. Hence

$$S_n f = \sum_{\|k\|_\infty \leq n} \left(\sum_{\ell \in \mathbb{Z}^d} \hat{f}_{k+(2n+1)\ell} \right) \psi_k, \quad (4.6)$$

that is, the grid cannot separate frequencies which differ by an integer multiple of $(2n+1)$ in every component, and the k -th coefficient of the interpolant collects all of them.

Now we are able to bound the aliasing error. By (4.6) and $1 + \|k\|_2^2 \leq 1 + dn^2 \leq 2dn^2$ for $\|k\|_\infty \leq n$ we have

$$\begin{aligned}
\|P_n f - S_n f\|_{H^r(\mathbb{T}^d)}^2 &= \sum_{\|k\|_\infty \leq n} (1 + \|k\|_2^2)^r \left| \sum_{\ell \in \mathbb{Z}^d \setminus \{0\}} \hat{f}_{k+(2n+1)\ell} \right|^2 \\
&\leq (2d)^r n^{2r} \sum_{\|k\|_\infty \leq n} \left| \sum_{\ell \in \mathbb{Z}^d \setminus \{0\}} (1 + \|k + (2n+1)\ell\|_2^2)^{-\frac{s}{2}} (1 + \|k + (2n+1)\ell\|_2^2)^{\frac{s}{2}} \hat{f}_{k+(2n+1)\ell} \right|^2.
\end{aligned}$$

Applying Cauchy–Schwarz inequality yields

$$\begin{aligned}
\|P_n f - S_n f\|_{H^r(\mathbb{T}^d)}^2 &\leq (2d)^r n^{2r} \sum_{\|k\|_\infty \leq n} \left(\sum_{\ell \in \mathbb{Z}^d \setminus \{0\}} (1 + \|k + (2n+1)\ell\|_2^2)^{-s} \right) \\
&\quad \left(\sum_{\ell' \in \mathbb{Z}^d \setminus \{0\}} (1 + \|k + (2n+1)\ell'\|_2^2)^s |\hat{f}_{k+(2n+1)\ell'}|^2 \right).
\end{aligned}$$

For the first inner sum we note that for $\|k\|_\infty \leq n$ and $\ell \neq 0$ every component satisfies

$$|k_i + (2n+1)\ell_i| \geq n|\ell_i|,$$

hence $\|k + (2n+1)\ell\|_2 \geq n\|\ell\|_2$ and $(1 + \|k + (2n+1)\ell\|_2^2)^{-s} \leq \|n\ell\|_2^{-2s} = n^{-2s} \|\ell\|_2^{-2s}$. Thus we are able to apply Lemma 12 and obtain

$$\begin{aligned}
&\|P_n f - S_n f\|_{H^r(\mathbb{T}^d)}^2 \\
&\leq \frac{2^{2d+1}s}{2s-d} (2d)^r n^{2r} n^{-2s} \sum_{\|k\|_\infty \leq n} \sum_{\ell' \in \mathbb{Z}^d \setminus \{0\}} (1 + \|k + (2n+1)\ell'\|_2^2)^s |\hat{f}_{k+(2n+1)\ell'}|^2 \\
&\leq \frac{2^{2d+1}s}{2s-d} (2d)^r n^{2(r-s)} \|f\|_{H^s(\mathbb{T}^d)}^2,
\end{aligned}$$

where the last step uses that $(k, \ell) \mapsto k + (2n+1)\ell$ is a bijection from $K_n \times \mathbb{Z}^d$ onto $\mathbb{Z}^d$, such that the sets $\{k + (2n+1)\ell : \ell \neq 0\}$, $k \in K_n$, are pairwise disjoint. Combining the inequalities for the projection error and the aliasing error yields

$$\|\text{Id} - S_n\|_{H^s(\mathbb{T}^d) \rightarrow H^r(\mathbb{T}^d)}^2 \leq \left(1 + \frac{2^{2d+1}s}{2s-d}\right) (2d)^r n^{2(r-s)} \leq \frac{2^{2d+2}s}{2s-d} (2d)^r n^{2(r-s)},$$

where we used $\frac{2^{2d+1}s}{2s-d} \geq 1$ in the last step. The bound on S_n follows from

$$\|S_n\|_{H^s \rightarrow H^s} \leq 1 + \|\text{Id} - S_n\|_{H^s \rightarrow H^s}$$

and the case $r = s$, since $2^{d+1} \sqrt{\frac{s}{2s-d}} (2d)^s \geq 1$. ■

Note, that the trigonometric interpolation operator S_n does not utilize the smoothness information, while the rate in the upper bound improves for higher smoothness. Hence, Fourier-based interpolation is universal in the sense that the error bounds adapt to the smoothness of the target function f without changing the implementation.

We further briefly comment on extensions to other Sobolev spaces.

Remark 14. *The above result holds for Sobolev spaces H_p^s with $1 < p < \infty$, where H_p^s is defined by replacing the L^2 -norm in (4.2) by L^p . It holds when the error norm is replaced by L^q with $1 < q < \infty$ as well. For $p, q \in \{1, \infty\}$ the same rate can be achieved using de la Vallée Poussin kernels, cf. [25, Theorem 3.6.4].*

4.4. Error bounds for the Koopman approximants. We are now prepared to state a main result of this work. Combining the approximation properties of trigonometric interpolation with the boundedness of the Koopman operator yields an L^2 error bound for the Fourier-EDMD approximation.

Theorem 15 (Error bound for Koopman approximation). *Let $\sigma \in \mathbb{N}$, $\sigma > d/2$ and assume that $F \in C^\sigma(\mathbb{T}^d)$ is a C^1 -diffeomorphism satisfying $c_0^{-1} := \inf_{x \in \mathbb{T}^d} |\det DF(x)| > 0$. Then for all $s \in (d/2, \sigma]$, the Koopman approximation $\hat{\mathcal{K}}_1$ defined in (3.2) satisfies the error bound*

$$\|\mathcal{K} - \hat{\mathcal{K}}_1\|_{H^s(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)} \leq C n^{-s}$$

with

$$C := 2^{d+1} \sqrt{\frac{s}{2s-d}} A(s), \quad A(s) = (2\pi)^s c_0^{\frac{\sigma-s}{2\sigma}} (\sigma!)^{\frac{s}{2\sigma}} C_F^{s/\sigma}.$$

The approximation $\hat{\mathcal{K}}_2$, defined in (3.3), satisfies the bound

$$\|\mathcal{K} - \hat{\mathcal{K}}_2\|_{H^s(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)} \leq \tilde{C} n^{-s}$$

with

$$\tilde{C} := 2^{d+1} \sqrt{\frac{s c_0}{2s-d}} + 2^{2d+3} \frac{s}{2s-d} (2d)^{s/2} A(s).$$

Proof. We abbreviate operator norms by dropping the domain, i.e. $\|\cdot\|_{H^s \rightarrow L^2} := \|\cdot\|_{H^s(\mathbb{T}^d) \rightarrow L^2(\mathbb{T}^d)}$. The proof follows [10, Theorem 3.4]. For the approximant $\hat{\mathcal{K}}_1 = S_n \mathcal{K}$, we have

$$\mathcal{K} - \hat{\mathcal{K}}_1 = \mathcal{K} - S_n \mathcal{K} = (\text{Id} - S_n) \mathcal{K}.$$

Substituting the bounds from Proposition 11 and Theorem 13 yields

$$\|\mathcal{K} - \hat{\mathcal{K}}_1\|_{H^s \rightarrow L^2} \leq \|\text{Id} - S_n\|_{H^s \rightarrow L^2} \|\mathcal{K}\|_{H^s \rightarrow H^s} \leq 2^{d+1} \sqrt{\frac{s}{2s-d}} A(s) n^{-s}.$$

For the approximant $\hat{\mathcal{K}}_2 = S_n \mathcal{K} S_n$, we have

$$\mathcal{K} - \hat{\mathcal{K}}_2 = \mathcal{K} - S_n \mathcal{K} S_n = \mathcal{K}(\text{Id} - S_n) + (\text{Id} - S_n) \mathcal{K} S_n.$$

Therefore, the approximation error satisfies

$$\begin{aligned} \|\mathcal{K} - \hat{\mathcal{K}}_2\|_{H^s \rightarrow L^2} &\leq \|\mathcal{K}(\text{Id} - S_n)\|_{H^s \rightarrow L^2} + \|(\text{Id} - S_n) \mathcal{K} S_n\|_{H^s \rightarrow L^2} \\ &\leq \|\mathcal{K}\|_{L^2 \rightarrow L^2} \|\text{Id} - S_n\|_{H^s \rightarrow L^2} + \|\text{Id} - S_n\|_{H^s \rightarrow L^2} \|\mathcal{K}\|_{H^s \rightarrow H^s} \|S_n\|_{H^s \rightarrow H^s} \end{aligned}$$

$$= (\|\mathcal{K}\|_{L^2 \rightarrow L^2} + \|\mathcal{K}\|_{H^s \rightarrow H^s} \|S_n\|_{H^s \rightarrow H^s}) \cdot \|\text{Id} - S_n\|_{H^s \rightarrow L^2}.$$

With $\|\mathcal{K}\|_{L^2 \rightarrow L^2} \leq \sqrt{c_0}$ from Theorem 8 with $\sigma = 0$, the interpolation error estimate from Theorem 13 as well as the uniform boundedness of S_n and $\mathcal{K}$ in $H^s(\mathbb{T}^d)$, the claim follows. $\blacksquare$

We compare this result with error bounds for kernel-based approximation.

Remark 16 (Comparison with fill-distance-based kEDMD bounds). *The error bounds of Theorem 15 rely crucially on the equispaced structure of the data set underlying trigonometric interpolation and thus differ from error bounds available for kernel EDMD (kEDMD), which apply to arbitrary data sets. In [10], deterministic L^∞ -error bounds for kEDMD are derived by identifying the regression problem with kernel interpolation in a reproducing kernel Hilbert space (RKHS) of Wendland functions, and bounding the resulting error in terms of the fill distance*

$$h_X := \sup_{x \in \mathbb{T}^d} \min_{x_i \in X} \|x - x_i\|_2$$

of an arbitrary data set $X = \{x_1, \dots, x_M\}$. For Wendland kernels of smoothness order $\tau \in \mathbb{N}_0$, the resulting convergence rate is of order $h_X^{\tau-d/2}$. For our equidistant grid of $(2n+1)^d$ points, the fill distance satisfies $h_X \asymp n^{-1}$, so that a rate of n^{-s} corresponds to the fill-distance rate h_X^s , rendering the error bound of Theorem 15 structurally similar to [10] in terms of the rate.

5. NUMERICAL EXAMPLE: KURAMOTO MODEL ON $\mathbb{T}^d$

As a scalable test case for Fourier-EDMD we use the *Kuramoto model* of coupled oscillators [14]. The model describes a system of d coupled oscillators, each characterized by a phase $\theta_i \in \mathbb{R}/2\pi\mathbb{Z}$, $i = 1, \dots, d$. Since every phase is a point on the circle, the natural state space for the joint system is the d -torus, $\mathbb{T}^d$. Each oscillator has an intrinsic natural frequency $\omega_i \in \mathbb{R}$, the rate at which it would rotate in the absence of coupling, and all oscillators are coupled via a *coupling strength* $\kappa \geq 0$ that pulls pairs of oscillators toward phase alignment. The classical, all-to-all coupled Kuramoto model is given by

$$\dot{\theta}_i = \nu_i + \frac{\kappa}{d} \sum_{j=1}^d \sin(\theta_j - \theta_i), \quad i = 1, \dots, d, \quad (5.1)$$

where the $1/d$ normalization of the coupling term is the standard scaling that keeps the right-hand side well behaved as $d \rightarrow \infty$. Rescaling $x_i = \theta_i/(2\pi) \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$ and applying one explicit-Euler step with step size $h > 0$ yields a discrete-time map $F : \mathbb{T}^d \rightarrow \mathbb{T}^d$,

$$F_i(x) = x_i + \omega_i + \frac{h\kappa}{2\pi d} \sum_{j=1}^d \sin(2\pi(x_j - x_i)) \pmod{1}, \quad i = 1, \dots, d. \quad (5.2)$$

The parameter $\omega_i := \frac{h\nu_i}{2\pi}$ describes the phase increment of the i -th uncoupled oscillator per time step. For $d = 1$ the coupling sum vanishes and (5.2) reduces to a rigid rotation. For the numerical implementation, the coupling sum is evaluated using the equivalent identity

$$\frac{1}{d} \sum_{j=1}^d \sin(2\pi(x_j - x_i)) = \frac{1}{d} \left(\cos(2\pi x_i) \sum_{j=1}^d \sin(2\pi x_j) - \sin(2\pi x_i) \sum_{j=1}^d \cos(2\pi x_j) \right), \quad (5.3)$$

replacing the $d \times d$ coupling matrix by two scalar sums per sampling point. This reduces both work and memory of evaluation of F from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.

We briefly verify that F satisfies the assumptions of Theorem 15. Clearly, $F \in C^\infty(\mathbb{T}^d, \mathbb{T}^d)$. Writing $F(x) = x + \omega + h\kappa g(x) \pmod{1}$ with $g_i(x) = \frac{1}{2\pi d} \sum_{j=1}^d \sin(2\pi(x_j - x_i))$, $i = 1, \dots, d$,

we have $\partial_{x_j} g_i = \frac{1}{d} \cos(2\pi(x_j - x_i))$ for $j \neq i$ and $\partial_{x_i} g_i = -\frac{1}{d} \sum_{j \neq i} \cos(2\pi(x_j - x_i))$ for all i , such that every row of $Dg(x)$ has absolute row sum at most $2(d-1)/d$. Hence

$$\|h\kappa Dg(x)\|_\infty \leq \frac{2h\kappa(d-1)}{d} < 2h\kappa < 1 \quad \text{for } h\kappa < \frac{1}{2},$$

uniformly in x , in the maximum row sum norm. Consequently the lift $x \mapsto x + \omega + h\kappa g(x)$ of F to $\mathbb{R}^d$ is the identity plus a contraction. Therefore, it is bijective on $\mathbb{R}^d$ and $DF = I + h\kappa Dg$ is invertible. Since g is 1-periodic, F is a C^∞ -diffeomorphism of $\mathbb{T}^d$. Moreover, all eigenvalues μ of $h\kappa Dg(x)$ satisfy $|\mu| \leq \|h\kappa Dg(x)\|_\infty < 2h\kappa$ and thus $|1 + \mu| > 1 - 2h\kappa$. Therefore

$$\inf_{x \in \mathbb{T}^d} |\det DF(x)| \geq (1 - 2h\kappa)^d,$$

Remark 17. The condition $h\kappa < \frac{1}{2}$ is equivalently given by $h < 1/(2\kappa)$ and hence only restricts the step size and not the model. This is what one expects, since the flow map of (5.1) is a diffeomorphism for every κ and the explicit-Euler map inherits this property as soon as the step size is chosen sufficiently small. Moreover $c_0 \leq (1 - 2h\kappa)^{-d} \rightarrow 1$ as $h \rightarrow 0$.

The phase increments ω_i are drawn i.i.d. from $\text{Unif}(0.05, 0.15)$ to avoid resonances between oscillators, and the coupling per time step is fixed at $h\kappa = 0.3$, which satisfies $h\kappa < 1/2$.

As observable to illustrate the error bound numerically, we choose a d -dimensional tensor-product B-spline of order two, $f : \mathbb{T}^d \rightarrow \mathbb{R}$, given by

$$f(x) = \prod_{i=1}^d \sqrt{12} \min\{x_i, 1 - x_i\}. \quad (5.4)$$

The normalization is chosen such that $\|f\|_{L^2(\mathbb{T}^d)} = 1$ for all d . The observable is continuous but not differentiable at $x_i = 1/2$ and, as a 1-periodic function, at $x_i = 0$. To be precise, it satisfies $f \in H^s(\mathbb{T}^d)$ for all $s < \frac{3}{2}$, but $f \notin H^{3/2}(\mathbb{T}^d)$. Since the convergence rate in Theorem 15 of the Koopman approximation depends on the regularity of the observable this example allows us to assess whether the numerically observed rate coincides with the theoretically predicted one.

Remark 18. Theorem 13 and consequently Theorem 15 require the observable to have Sobolev smoothness $s > d/2$. For the B-spline of order two in (5.4), this restricts the applicability of the theory to $d \leq 2$. Nevertheless, we present the numerical results up to $d = 5$, since the observed convergence rates are unaffected. Still, this condition cannot be removed from our worst-case analysis. It ensures both the convergence of the series in Lemma 12 and the continuous embedding $H^s(\mathbb{T}^d) \hookrightarrow C(\mathbb{T}^d)$. For $s \leq d/2$ point evaluation is unbounded on $H^s(\mathbb{T}^d)$ and hence no algorithm using finitely many point evaluations can have a finite worst-case error on this space. However, this does not contradict the observed rates, since we do not consider a worst-case error over $H^s(\mathbb{T}^d)$, but the error for one particular observable. Since our observable (5.4), the dictionary, and the grid X have a tensor-product structure, the interpolation operator also factorises. Hence, the d -dimensional interpolation error is governed by the corresponding one-dimensional ones, for which only $s > 1/2$ is required.

In the following, we describe the implementation of the two Fourier-EDMD approximations $\hat{\mathcal{K}}_1$ and $\hat{\mathcal{K}}_2$ of the Koopman operator and then investigate their convergence behavior as well as the computational cost. For this, for a given bandwidth n we use the equispaced tensor-product grid X defined in (2.7), i.e., the grid contains $N = (2n+1)^d$ grid points and N Fourier frequencies. Both approximations $\hat{\mathcal{K}}_1$ and $\hat{\mathcal{K}}_2$ are built on this grid. They only differ in the way the composition of the observable with the dynamics is evaluated, depending on which data are assumed to be available.

For the approximation $\hat{\mathcal{K}}_1 = S_n \mathcal{K}$ defined in (3.2), the values $f(F(x_j))$ are assumed to be given, i.e. the composition $f \circ F$ is evaluated directly on the equispaced grid. More precisely, the observable

is sampled at image points $F(x_j)$ and a single d -dimensional FFT returns the corresponding discrete Fourier coefficients $\widehat{(f \circ F)}_k$ of the Koopman image. Neither the matrix L_X nor L_{X+} is ever explicitly formed. Since the number of grid points $N = (2n + 1)^d$ increases exponentially in d , a direct construction of the full $d \times N$ coordinate array would become extremely expensive in higher dimensions. Therefore, we evaluate the observable and the dynamics blockwise along one coordinate direction, using the representation (5.3) in each block. This reduces the size of the temporary coordinate arrays from $\mathcal{O}(dN)$ to $\mathcal{O}(dB(2n + 1)^{d-1})$, where the block size B is chosen from a prescribed memory budget. However, the function values as well as the corresponding Fourier coefficients still require $\mathcal{O}(N)$ storage.

In contrast to $\widehat{\mathcal{K}}_1$, for $\widehat{\mathcal{K}}_2 = S_n \mathcal{K} S_n$ defined in (3.3), only point evaluations $f(x_j)$ of the observable on the uniform grid are available. Hence, we first replace f by its interpolant $S_n f$ and then evaluate the Fourier representation of $S_n f$ directly at image points $F(x_j)$,

$$(S_n f)(F(x_j)) = \sum_{k \in K_n} \tilde{f}_k \exp(2\pi i \langle k, F(x_j) \rangle).$$

This corresponds exactly to the multiplication with the matrix L_{X+} discussed after (3.3). Since the points $F(x_j)$ are in general not equispaced, this evaluation is carried out by a type-2 nonuniform FFT (NFFT/NUFFT), see [21]. The resulting values are then transformed back by a second d -dimensional FFT. For the NFFT step we use the NFFT3 library [8] through its Python interface pyNFFT3¹. We set the cut-off $m = 7$ of the window function and the oversampling factor $\sigma \geq 2$. For these parameters, the relative error of the NFFT stays below 10^{-12} , i.e. the approximation error is not influenced by the NFFT. Although the NFFT3 library is not restricted to a particular dimension, the approximant $\widehat{\mathcal{K}}_2$ is illustrated only for $d \leq 3$. The cost of one NFFT contains a factor $(2m + 2)^d$ stemming from the support of the window function. Consequently, for $d \geq 4$ the NFFT dominates the overall cost and $\widehat{\mathcal{K}}_2$ is no longer computationally practical at the bandwidths considered here. As for $\widehat{\mathcal{K}}_1$, the points $F(x_j)$ are generated blockwise, such that the full $d \times N$ coordinate array is never stored simultaneously. In summary, the computation of $\widehat{\mathcal{K}}_1$ requires only one FFT and one evaluation of the composition $f \circ F$ on X , whereas $\widehat{\mathcal{K}}_2$ requires two FFTs, one evaluation of the dynamics F on the grid and one NFFT. That is, $\widehat{\mathcal{K}}_2$ costs one NFFT and one FFT more than $\widehat{\mathcal{K}}_1$.

In order to quantify the convergence, we consider the relative L^2 -error

$$\frac{\|\mathcal{K}f - \widehat{\mathcal{K}}_i f\|_{L^2(\mathbb{T}^d)}}{\|\mathcal{K}f\|_{L^2(\mathbb{T}^d)}} \quad i = 1, 2 \quad (5.5)$$

since we compare the approximation results across different dimensions d . We estimate both norms using Monte-Carlo quadrature

$$\|\mathcal{K}f - \widehat{\mathcal{K}}_i f\|_{L^2(\mathbb{T}^d)} \approx \left(\frac{1}{M} \sum_{m=1}^M |f(F(z_m)) - \widehat{\mathcal{K}}_i f(z_m)|^2 \right)^{1/2},$$

where $z_1, \dots, z_M$ are drawn independently and uniformly from $\mathbb{T}^d$. Hence, the error is evaluated independently of the grid X that was used to construct the Koopman approximation and in particular does not vanish at the interpolation nodes. Evaluating $\widehat{\mathcal{K}}_i f$, $i = 1, 2$ at M arbitrary points costs $\mathcal{O}(MN)$, i.e. M would have to be reduced as the dictionary grows. This can be avoided by performing this evaluation with type-2 NFFT as well, at cost $\mathcal{O}(N \log N + M)$ with the same NFFT parameters as above. With this, we can choose up to dimension $d = 3$ and for all bandwidths $M = 10^5$ test points. For $d \geq 4$ the error is evaluated directly and M is chosen between $2 \cdot 10^3$ and $5 \cdot 10^3$ so that MN stays bounded.

¹see <https://pypi.org/project/pyNFFT3/>

Figure 1 shows the relative L^2 -errors (5.5) of the approximants $\hat{\mathcal{K}}_1$ and $\hat{\mathcal{K}}_2$ for the B-spline observable (5.4). Since $f \in H^s(\mathbb{T}^d)$ for all $s < 3/2$, Theorem 15 predicts the rate n^{-s} for every $s < 3/2$. In the top row, the error decay in terms of the bandwidth n is illustrated. The five curves are parallel showing that the predicted rate n^{-s} , $s < 3/2$, is attained independently of d . The measured rates lie between -1.46 and -1.56 in all dimensions $d = 1, \dots, 5$ (see Table 1) and for both approximants, i.e. they attain the predicted rate up to the accuracy of the Monte-Carlo error estimate. Notably this includes $d \geq 3$, where the assumption $s > d/2$ of Theorem 13 is violated (Remark 18). Expressed in terms of the total number $N = (2n + 1)^d$ of samples, the same data give the rate $N^{-s/d}$ (bottom row of Figure 1), which is the familiar curse of dimensionality for isotropic Sobolev smoothness.

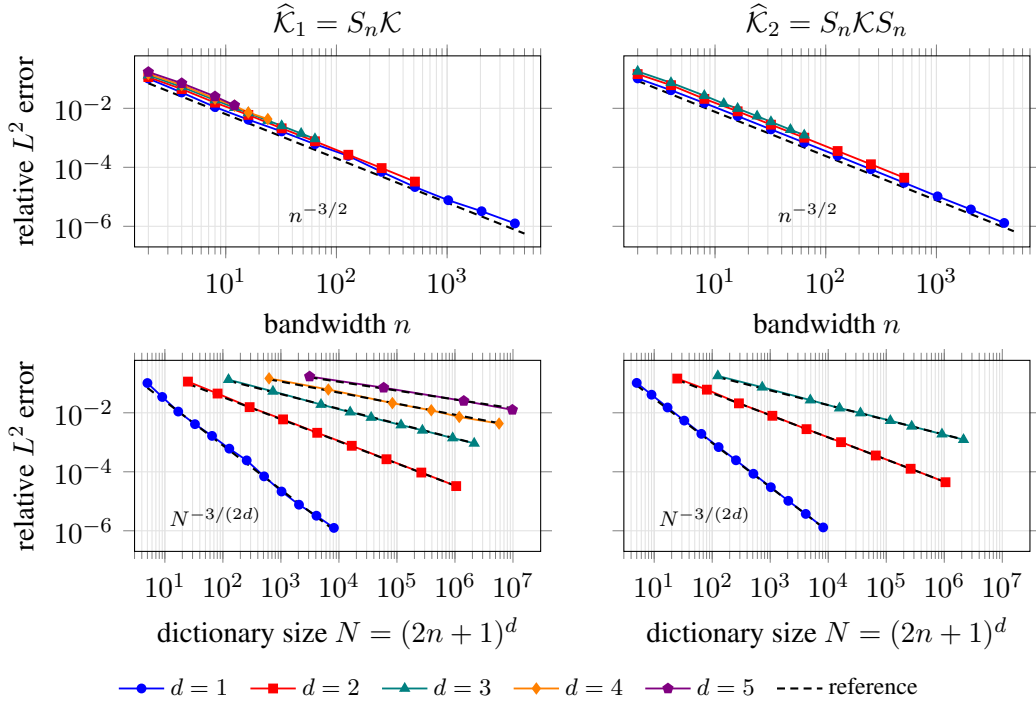


FIGURE 1. Relative L^2 -error of the two Koopman approximants for the observable (5.4) and the Kuramoto map (5.2); left column $\hat{\mathcal{K}}_1$, right column $\hat{\mathcal{K}}_2$ (available for $d \leq 3$). Top row: in terms of the bandwidth n ; Bottom row: in terms of the number $N = (2n + 1)^d$ of samples.

Table 1 makes the comparison of the observed convergence rate with the theoretical one quantitative. It shows, for each dimension and each approximant, the decay rate of the measured error against the bandwidth in a log-log fit. For both approximants and all bandwidths n , the rates lie between -1.46 and -1.56 against the predicted $-s$ with $s < 3/2$. Moreover, the last column provides information that is not visible in Figure 1 due to scaling. It gives the range of the ratio $\|\mathcal{K}f - \hat{\mathcal{K}}_2 f\|_{L^2} / \|\mathcal{K}f - \hat{\mathcal{K}}_1 f\|_{L^2}$. This ratio stays bounded instead of growing with n , and shows in particular that the rates of the two approximants differ only by a constant factor. This is exactly the behavior predicted by Theorem 15, in which $\hat{\mathcal{K}}_1$ and $\hat{\mathcal{K}}_2$ have the same rate n^{-s} and differ only in the constants C and $\tilde{C}$.

Figure 2 shows the corresponding computation times. By the discrete Fourier structure of L_X established in Section 2, both approximants are expected to cost $\mathcal{O}(N \log N)$. The top row shows the growth in the bandwidth n , the bottom row the same data against the dictionary size N . The reference lines are drawn only over the range in which the runtime follows the asymptotic behavior and is not

TABLE 1. Observed convergence rates for the observable (5.4).

d	bandwidth	$N_{\max}$	rate $\hat{\mathcal{K}}_1$	rate $\hat{\mathcal{K}}_2$	error ratio
1	$8 \leq n \leq 4096$	8 193	-1.49	-1.50	1.01–1.40
2	$8 \leq n \leq 512$	1 050 625	-1.49	-1.48	1.27–1.36
3	$8 \leq n \leq 64$	2 146 689	-1.46	-1.48	1.33–1.42
4	$8 \leq n \leq 24$	5 764 801	-1.48	–	–
5	$4 \leq n \leq 12$	9 765 625	-1.56	–	–

dominated by the constant overhead. They are of the form $C N \log N$, in the top row with one constant per dimension and in the bottom row with a single constant for all dimensions. For $\hat{\mathcal{K}}_1$ this confirms that the cost is governed by the dictionary size, not by the dimension. However, for $\hat{\mathcal{K}}_2$ the constant grows with d , again through the window function of the NFFT. For the largest bandwidth in each dimension $d \leq 3$, building $\hat{\mathcal{K}}_2$ is a factor of 3.5 (for $d = 1, 2$) to 10.3 (for $d = 3$) more expensive than $\hat{\mathcal{K}}_1$, reflecting the NFFT and the additional FFT it performs. The largest dictionary, $d = 5$ and $n = 12$, uses $N = 25^5 = 9\,765\,625$ Fourier modes and takes 1.4 s. The corresponding EDMD matrix has $9.5 \cdot 10^{13}$ entries, about 1.5 PB in complex double precision, whereas the matrix-free evaluation stores a coefficient array of 149 MB. That this remains feasible in higher dimensions is due to the blockwise evaluation of the dynamics and the observable, which never stores the full $d \times N$ coordinate array.

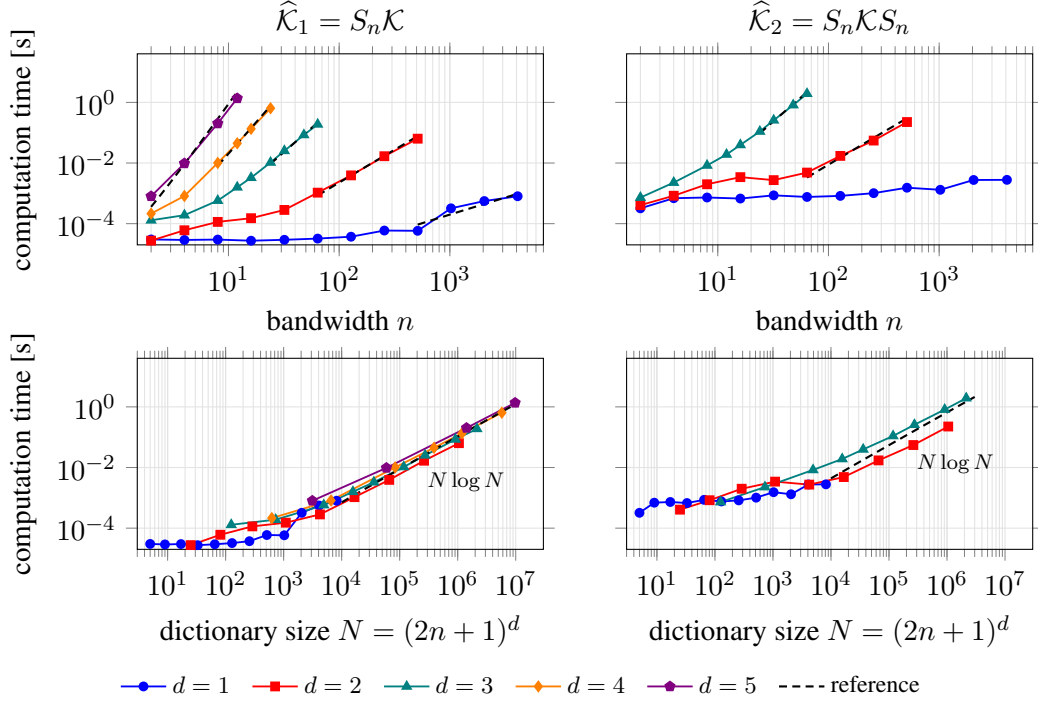


FIGURE 2. Computation time for constructing the two Koopman approximants excluding the error estimation; same data, colors and layout as in Figure 1.

6. CONCLUSION

We have studied extended dynamic mode decomposition with a Fourier dictionary on the torus, where the data are sampled on an equispaced tensor grid whose cardinality matches the dimension of

the dictionary. To this end, we first showed that the EDMD least-squares problem admits a unique closed-form solution, the inverse of the Fourier sampling matrix being, up to scaling, its adjoint. This allowed us to identify the EDMD matrix with the matrix representation of trigonometric interpolation composed with the Koopman operator. That is, Fourier-EDMD on equispaced grids coincides with trigonometric interpolation of the Koopman image. Based on this identification, we derived error bounds of optimal order n^{-s} in the bandwidth n for observables of Sobolev smoothness $s > d/2$, both for the approximant which uses measurements of the observable along the dynamics and for the fully data-driven approximant, with constants that are explicit in terms of the map F , its smoothness, the dimension d and s . Along the way, we provided an error estimate for trigonometric interpolation with explicit constants, boundedness of the Koopman operator on Fourier-based Sobolev spaces of fractional order and the isometric identification of these spaces with interpolation spaces. Moreover, the same structure yields a matrix-free implementation by means of the FFT and, for the nonequispaced image points, the NFFT, with quasi-linear cost in the dictionary size. We illustrated the results for the Kuramoto model on $\mathbb{T}^d$ up to $d = 5$ with dictionaries of roughly 10^7 modes, where the predicted rates were observed, for $d \geq 3$ even beyond the range $s > d/2$ covered by our analysis, and the cost was confirmed to be quasi-linear in the dictionary size.

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